Experiment 6

Dashboard and Scorecard Development Using Microsoft Power BI

AIM:

To design and develop an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI for monitoring organizational performance, supporting business decision-making, and communicating key performance indicators (KPIs) effectively.

OBJECTIVES:

- To understand the importance of dashboards and scorecards in Business Intelligence.
- To design an interactive Executive Dashboard using Power BI.
- To create KPI Cards for Total Sales, Total Profit, Total Orders, and Profit Margin.
- To visualize sales and profit performance using different charts.
- To implement interactive slicers for Region, Category, and Year.
- To create a Business Scorecard for comparing target and actual performance.
- To apply performance status indicators such as Achieved, Partially Achieved, and Not Achieved.
- To analyze business performance and identify important trends.
- To develop data-driven business recommendations.

TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

DATASET USED: Global Superstore Dataset.xlsx. https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

THEORY:

Dashboard: A dashboard is a visual interface that presents important business metrics, trends, and performance indicators in a consolidated view. It helps managers and executives monitor organizational performance and make informed decisions.

Business Scorecard: A business scorecard is a performance-monitoring tool used to compare actual business results against predefined targets. It helps organizations evaluate progress toward their objectives.

Key Performance Indicators (KPIs): KPIs are measurable values used to evaluate the performance of an organization or business process. Examples include Total Sales, Total Profit, Total Orders, and Profit Margin.

Power BI: Microsoft Power BI is a Business Intelligence tool used to connect, analyze, and visualize data through reports, dashboards, charts, tables, and interactive filters.

DAX: Data Analysis Expressions (DAX) is a formula language used in Power BI to create measures and calculate business metrics such as profit margin, target performance, and order counts.

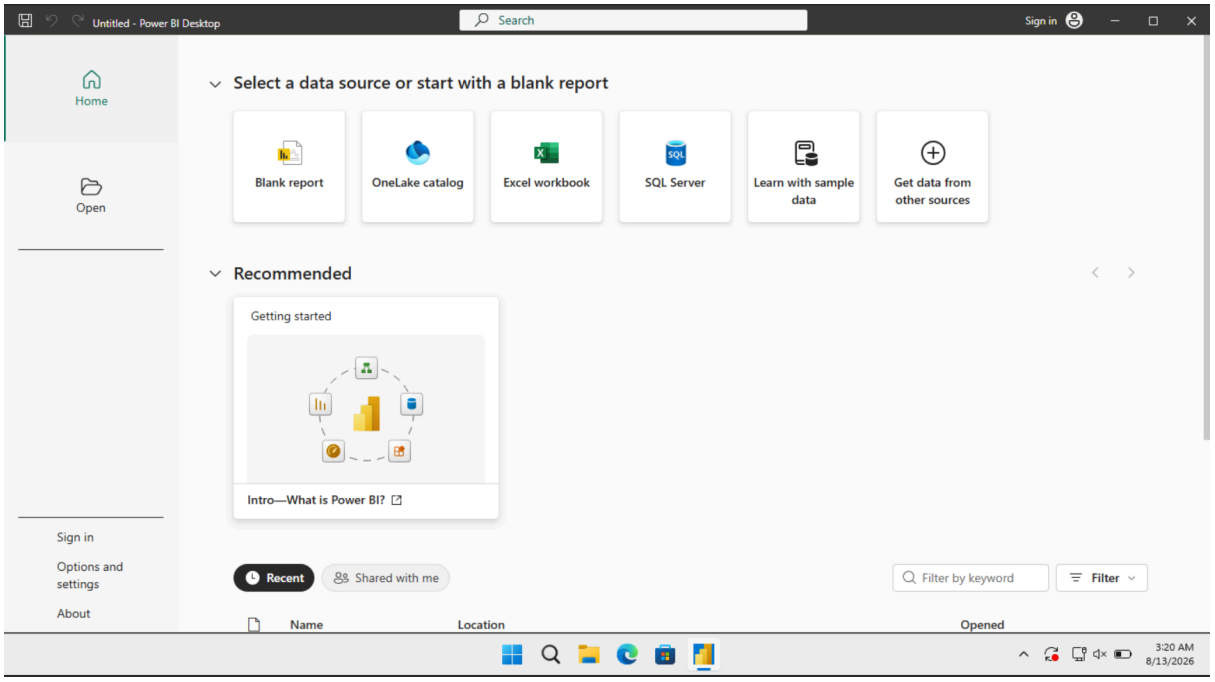
Slicers: Slicers are interactive visual filters that allow users to filter report data according to fields such as Region, Category, and Year.

Conditional Formatting: Conditional formatting uses colors or other visual indicators to highlight performance status and make comparisons easier.

PROCEDURE:

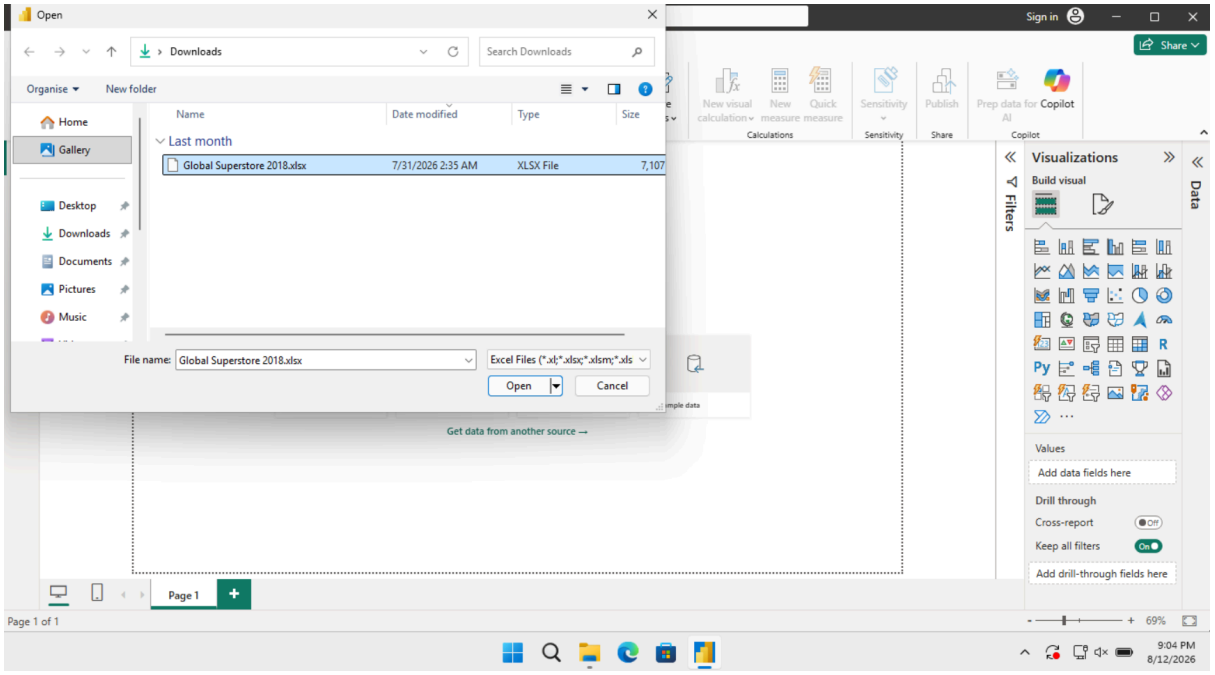
Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.



Step 2: Import the Global Superstore dataset.

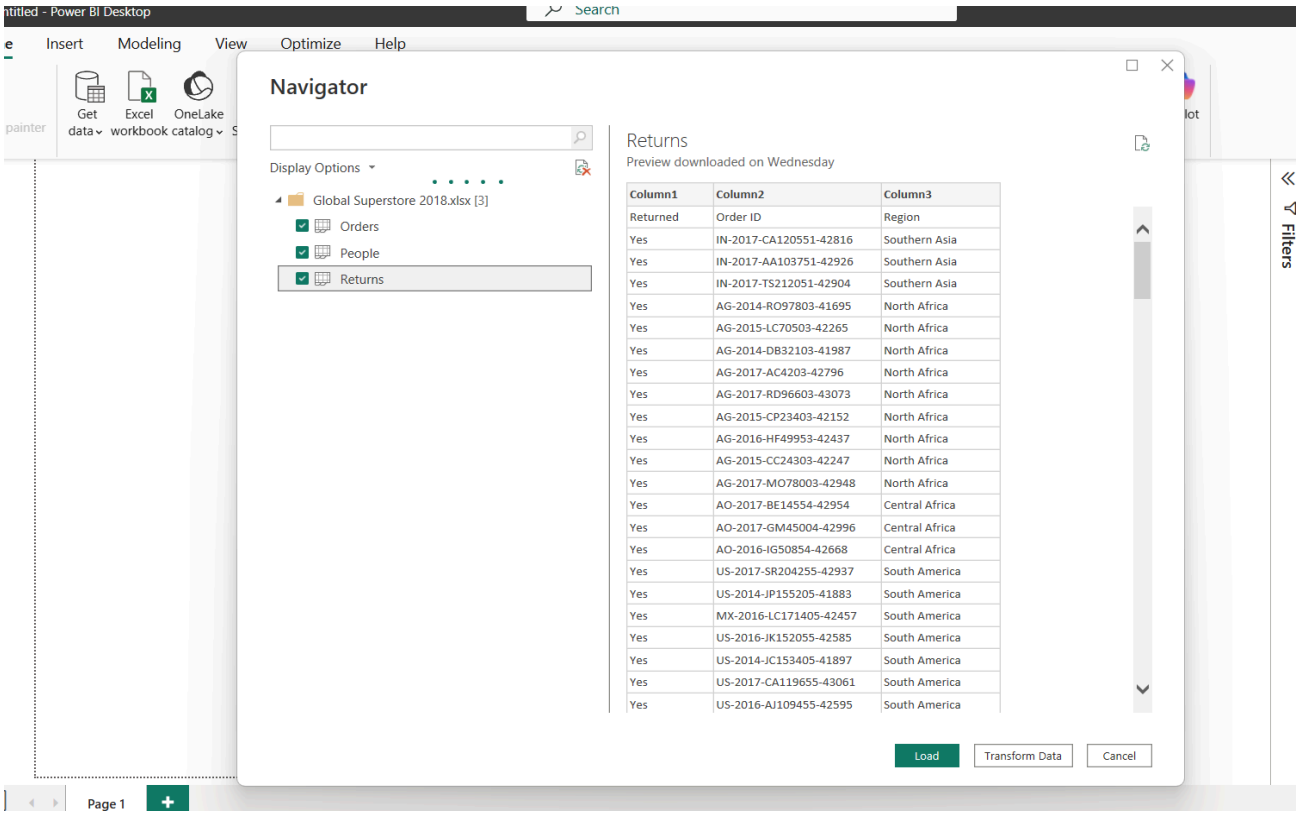
Home > Get Data > Excel Workbook Select:



Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:



• Orders

• Returns

• People

Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables

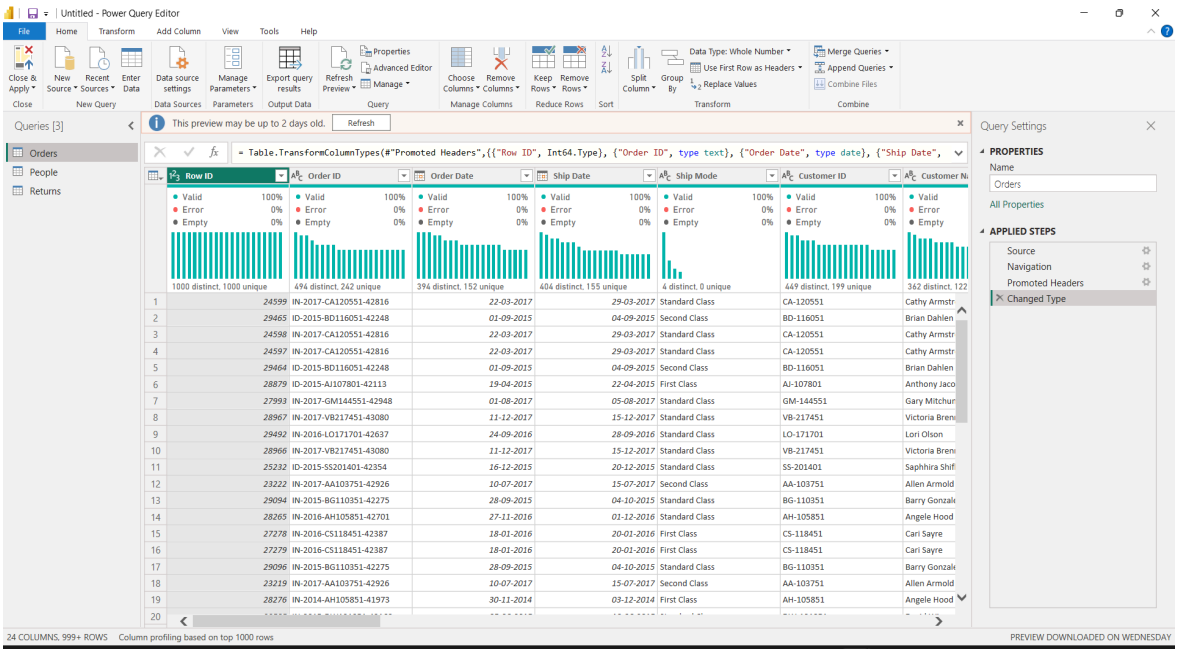
Step 4: Analyse the Orders table.

Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity
- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.



Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data. Organize the attributes into:

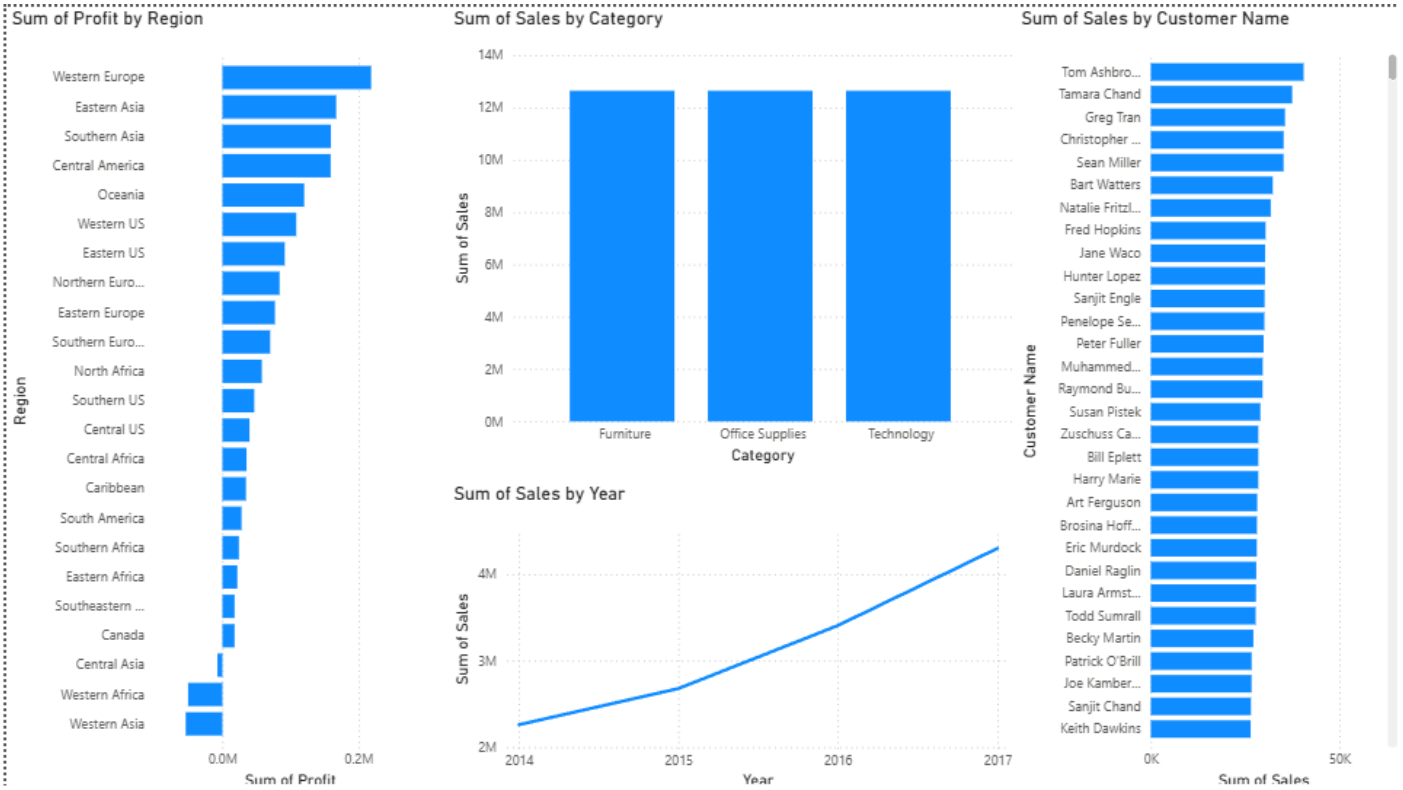
A. Fact Table: Fact_Sales

B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

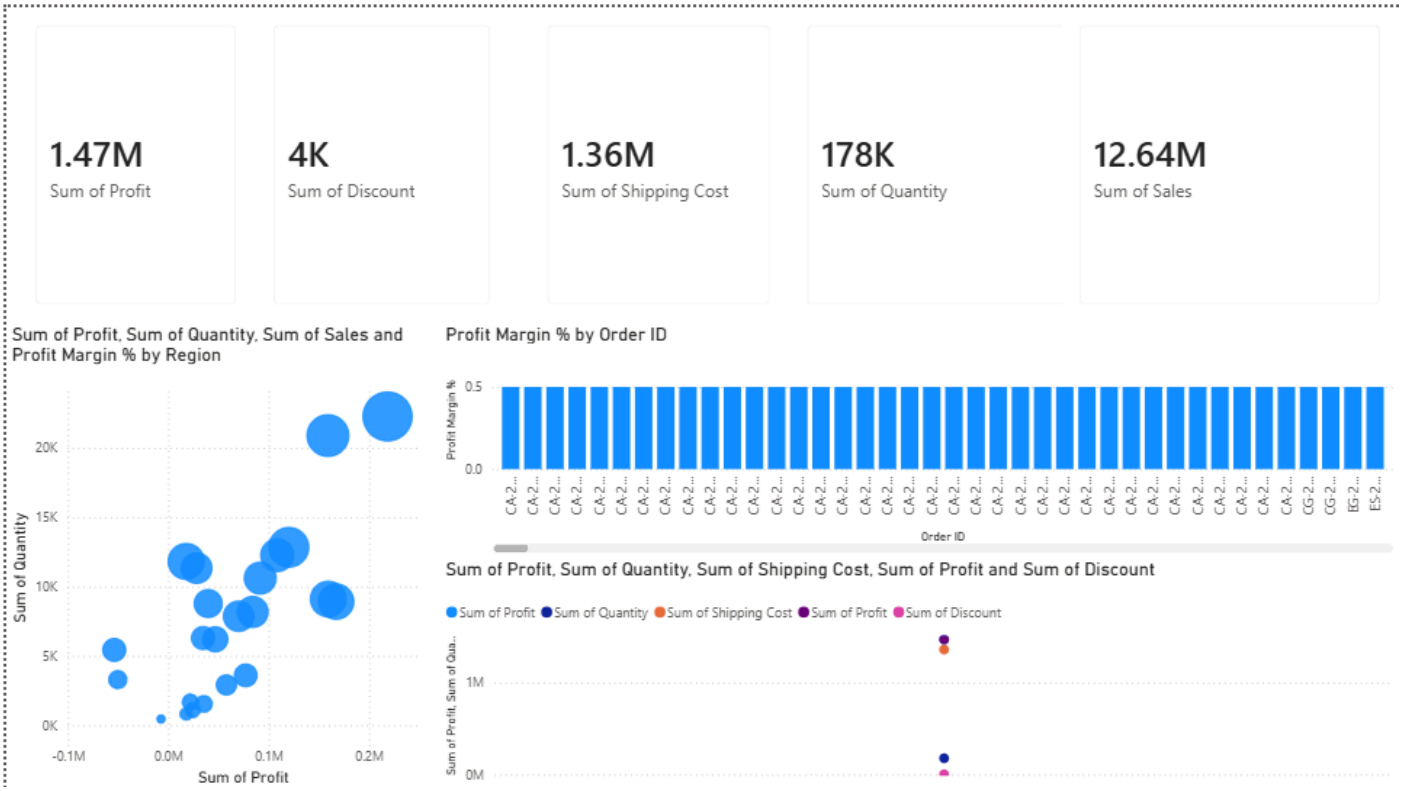
This classification forms the basis of the Star Schema.

Part B: Dashboard Planning

1. Created a new report page named Executive Dashboard.
2. Planned the dashboard layout by arranging KPI Cards, charts, slicers, and the scorecard in a structured manner.
3. Selected suitable visualizations for presenting business performance indicators.



Part C: Creating the KPI Dashboard



Step 1: Total Sales Card

Created a Card visual and added the Sales field with Sum aggregation to display total sales.

Step 2: Total Profit Card

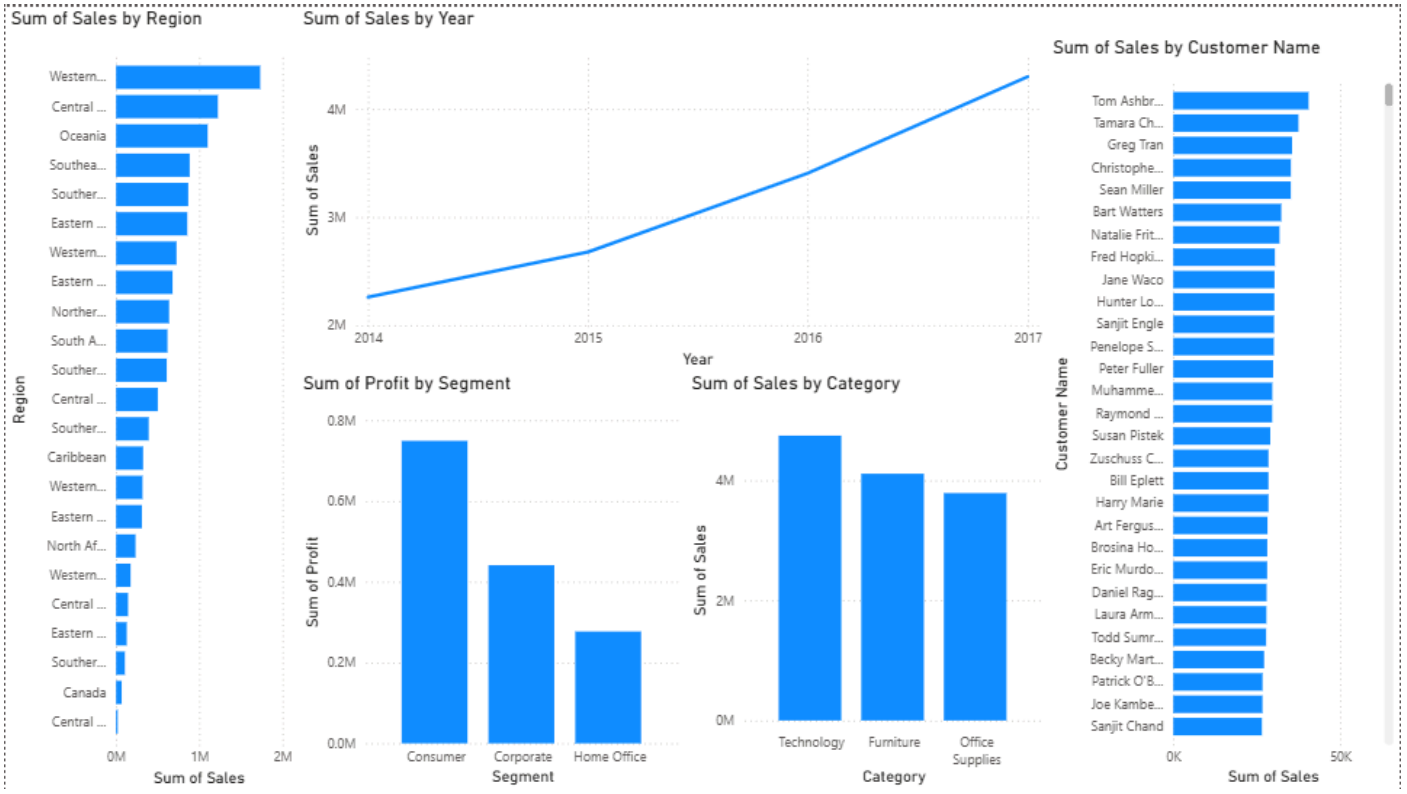
Created a Card visual and added the Profit field with Sum aggregation to display total profit.

Step 3: Total Orders Card

Created a Card visual and used the Order ID field with Distinct Count aggregation to count unique orders.Step 4: Profit Margin Card

Created a DAX measure to calculate the percentage of profit earned relative to sales.

```
Profit Margin % =
DIVIDE(
  SUM(Fact_Sales[Profit]),
  SUM(Fact_Sales[Sales]),
  0
)
```



Formatted the measure as a percentage and displayed it using a Card visual.

Step 5: Sales by Region

Created a Clustered Bar Chart by placing Region on the Y-axis and Sum of Sales on the X-axis.

Step 6: Sales by Category

Created a Clustered Column Chart by placing Category on the X-axis and Sum of Sales on the Y-axis.

Step 7: Monthly Sales Trend

Created a Line Chart using Order Date on the X-axis and Sum of Sales on the Y-axis. Used the date hierarchy to analyze sales over time at the monthly level.

Step 8: Top 10 Customers by Sales

Created a Clustered Bar Chart using Customer Name and Sum of Sales. Applied a Top N visual-level filter to display the ten customers with the highest sales.

Step 9: Profit by Segment

Created a Clustered Column Chart using Segment on the X-axis and Sum of Profit on the Y-axis.

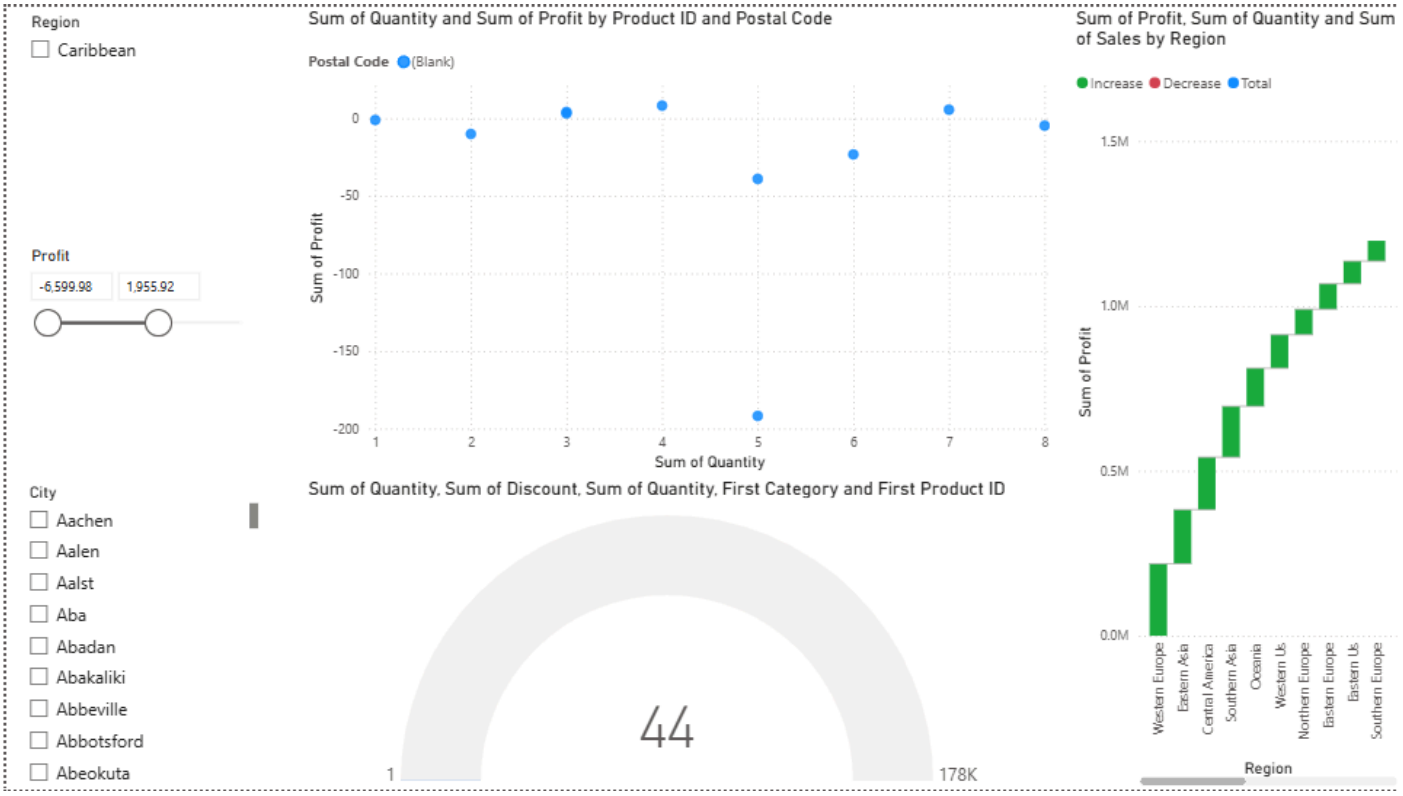
Step 10: Formatting the Dashboard

Added meaningful titles and arranged the KPI Cards and charts to make the dashboard easy to read and interpret.

Part D: Adding Interactive Features

Step 1: Region Slicer

Added a slicer using the Region field and assigned the title Filter by Region.



Step 2: Category Slicer

Added a slicer using the Category field and assigned the title Filter by Category.

Step 3: Year Slicer

Added a slicer based on the Year level of the Order Date hierarchy and assigned the title Filter by Year.

Step 4: Testing Interactivity

Selected different regions, categories, and years to verify that the KPI Cards and charts responded to the selections. Checked the visual interactions to ensure that the report remained interactive.

Part E: Creating the Business Scorecard

Step 1: Sales Target

Created a DAX measure to define a sample sales target.

Sales Target = 1200000

Created a Sales Status measure to compare actual sales against the target.

```
Sales Status =
VAR ActualSales = SUM(Fact_Sales[Sales])
VAR TargetSales = [Sales Target]
RETURN
  SWITCH(
    TRUE(),
    ActualSales >= TargetSales, "Achieved",
    ActualSales >= TargetSales * 0.8, "Partially Achieved",
    "Not Achieved"
)
```

Step 2: Profit Target

Created a sample profit target measure.

Profit Target = 150000

Created a Profit Status measure to evaluate profit performance.

```
Profit Status =
VAR ActualProfit = SUM(Fact_Sales[Profit])
VAR TargetProfit = [Profit Target]
RETURN
  SWITCH(
    TRUE(),
    ActualProfit >= TargetProfit, "Achieved",
    ActualProfit >= TargetProfit * 0.8, "Partially Achieved",
    "Not Achieved"
)
```

Step 3: Order Target

Created a sample order target measure.

```
Order Target = 5000
```

Created an Order Status measure using the distinct count of Order ID.

```
Order Status =
VAR ActualOrders = DISTINCTCOUNT(Fact_Sales[Order ID])
VAR TargetOrders = [Order Target]
RETURN
  SWITCH(
      TRUE(),
      ActualOrders >= TargetOrders, "Achieved",
      ActualOrders >= TargetOrders * 0.8, "Partially Achieved",
      "Not Achieved"
)
```

Step 4: Creating the Scorecard Table

Created a Table visual and added the following fields:

- Sales Target
- Sum of Sales
- Sales Status
- Profit Target
- Sum of Profit
- Profit Status
- Order Target
- Distinct Count of Order ID
- Order Status

Part F: Business Analysis

1. Examined the Sales by Region chart to compare regional sales performance.
2. Examined the Sales by Category chart to compare the sales generated by different product categories.
3. Used the Monthly Sales Trend chart to observe changes in sales over time.
4. Analyzed the Top 10 Customers chart to identify customers contributing the highest sales.
5. Compared profit across Consumer, Corporate, and Home Office segments.
6. Reviewed the Business Scorecard to evaluate actual performance against the sample targets.

7. Identified the best-performing and worst-performing regions and the most and least profitable categories based on the displayed results.
8. Considered business recommendations based on the observed performance patterns.

OBSERVATIONS

1. KPI Cards provided a consolidated view of Total Sales, Total Profit, Total Orders, and Profit Margin.
2. The Sales by Region chart enabled comparison of sales across geographical regions.
3. The Sales by Category chart showed the contribution of different product categories to overall sales.
4. The Monthly Sales Trend chart helped examine changes in sales over time.
5. The Top 10 Customers chart highlighted the customers with the highest sales contribution.
6. The Profit by Segment chart supported comparison of profitability among customer segments.
7. Region, Category, and Year slicers enabled interactive filtering of report data.
8. The Business Scorecard displayed sample targets alongside actual values and performance statuses.
9. The status measures classified performance as Achieved, Partially Achieved, or Not Achieved according to the defined thresholds.
10. The combined dashboard and scorecard provided a consolidated view of business performance.

RESULT:

An interactive Executive Dashboard and Business Scorecard were successfully developed using Microsoft Power BI. The dashboard included KPI Cards, sales and profit visualizations, Top 10 Customer analysis, and interactive slicers. A Business Scorecard was also created to compare actual sales, profit, and order performance against sample targets using DAX measures and performance-status indicators. The experiment demonstrated how Power BI can support business performance monitoring, data visualization, and data-driven decision-making.

LEARNING OUTCOMES:

- Learned to design interactive dashboards using Power BI visualizations. (CO5-BT6)
- Learned to create KPI Cards for monitoring business performance.
- Learned to analyze sales and profit using bar charts, column charts, and line charts.
- Learned to implement interactive slicers for Region, Category, and Year.
- Learned to create DAX measures for calculating profit margin and performance status.
- Learned to develop a target-versus-actual Business Scorecard. (CO5-BT6)
- Understood how to use performance indicators to monitor business objectives.
- Learned to interpret dashboard findings and identify business trends. (CO3-BT4)
- Developed the ability to formulate recommendations based on analytical findings. (CO5-BT6)